ANLC-752 — Final Submission Summary

Bitcoin AML Monitoring Database · Elliptic Data Set · Alpesh Ranjan

What was planned

The Unit 1 question: how can transaction-level Bitcoin data be structured in an RDBMS to support efficient identification and monitoring of patterns associated with illicit activity? This is a data-management question, not a modelling one — success is measured by what the schema makes queryable, not by a model score.

Unit 8 delivered the validated database: four schemas (staging, core, analytics, governance), an ETL pipeline running as a single transaction with nine named data quality rules, a monitoring layer mapped to laundering typologies, and an analysis-preparation layer — all in SQL. The milestone deck's closing slide scoped five remaining items, listed below.

What was executed

The database was rebuilt from the authentic Elliptic files and every claim below was re-measured live on PostgreSQL 18.4, not carried over from the earlier run.

Required item	Status	Where it lives
Fit and evaluate a model	Done	reports/02_model_evaluation.md
Partition by period	Done	reports/03_partitioning_and_refresh.md
Incremental refresh strategy	Done	reports/03_partitioning_and_refresh.md
Access control	Done	reports/04_access_control.md
Elliptic++ actor layer	Not attempted	requires a separate dataset (see page 2)

Headline results

Measure	Result
Load, full pipeline (203,769 tx · 234,355 edges)	committed in 17.3 s; all 9 quality rules passed
Illicit vs licit counterparty adjacency	30.0% vs 2.4% — the 12x separation, reproduced
Illicit rate, periods 42 to 46	11.10% falling to 0.28% — the ~40x collapse, reproduced
Partition pruning (period-scoped query)	1 of 49 partitions scanned; 0.671 ms
Incremental vs full refresh	109 ms vs 4,284 ms (~39x)
Row-level security	loader sees the row; analyst sees 0 rows
Fitted model vs existing heuristic	PR-AUC 0.287 vs 0.211; wins at 5%+ review budget

The case study — what the database actually hands an investigator

Aggregate statistics do not answer the question a compliance officer asks, which is about one transaction: *why this one?* Transaction **139273997** (period 28) is unlabelled by Elliptic but sits downstream of 59 transactions Elliptic confirmed illicit — 74 across the full upstream chain, which terminates at three hops. Its shape is the finding: 59 edges in, **zero** out — a pure sink, the consolidation signature. Detection and the traceable reason are both plain SQL.

What the work exposed — reported, not smoothed over

Decomposing the alert queue's risk score against real data revealed three flaws in the scoring heuristic that had not previously been visible:

1. The exposure term saturates. It caps at 5 illicit neighbours, so a transaction with 59 confirmed-illicit inputs scores identically to one with 6. The queue cannot tell them apart.

2. The top of the queue is a tie, not a ranking. All ten top rows score exactly 13.000, so their order is whatever the planner emits.

3. The top is structurally biased toward sinks. All ten have zero out-degree, though only 203 of 4,027 queued transactions (5.0%) are sinks — reaching a ratio of 1.0000 requires every neighbour to be illicit, and outgoing edges are extra neighbours that can dilute it. One term rewards fan-out while another quietly filters the top down to transactions with none.

This is a more useful outcome than a queue that had merely looked correct, and it is the concrete version of a caveat the view's own comment already carried: the weights are a stated heuristic, not fitted coefficients.

Honest limitations

Limitation	Consequence
Features are obfuscated	findings stateable as "feature 53 separates the classes", never in business terms
Edges join transactions, not parties	no customer dimension; nothing here identifies who did anything
Labels are binary and pre-2019	money laundering not separable from other illicit categories
Model not benchmarked	a logistic regression on 8 graph predictors; the 165 supplied features were not given to it, and published results on this dataset were not compared against
Actor layer not attempted	needs Elliptic++'s address-level files, a separate release not held here

Conclusion

The milestone delivered a validated database. The work that followed put it under the four pressures a real deployment applies — a model consuming it, a query pattern dominant enough to demand partitioning, a monitoring layer needing to stay current without a full rebuild, and more than one class of user needing different access — and it held under all four, verified live rather than asserted.

The strongest evidence for the original thesis is not the fitted model, which is deliberately modest. It is that a self-join and a GROUP BY — no model, no feature engineering — surfaced both the 12x adjacency separation and the 40-fold market collapse, using structure the three source CSVs do not contain on their own. Structuring the data did not merely make analysis faster; it produced the interpretable signal the raw data could not supply.

Where the detail lives

This summary is deliberately short. Every claim above traces to a document in the submission, and every figure to a query that can be re-run:

File	Contents
reports/01_overview.md	overview and status of all five items
reports/05_case_study_alert_triage.md	flagging rule, score decomposition, the three flaws, full trace
reports/02_model_evaluation.md	precision/recall at fixed review budgets, coefficients, limitations
reports/03_partitioning_and_refresh.md	migration trade-offs, EXPLAIN output, refresh timings
reports/04_access_control.md	role design and the live row-level security test
validation_run_log_rerun.md	raw terminal output of the second run, end to end
validation_run_log.txt	the original Unit 8 run, left exactly as generated
README.md · docs/data_dictionary.md	setup and reproduction · catalog reference, regenerated